

SUMMARY OF GUIDELINES:

Overview

- **Theme:** Traffic incidents in Chicago (2016–2024)
- **Goal:** Simulate a decision support system for an insurance company using data mining tools (Python)
- **Main Files:**
 - crashes.csv: Core data about incidents
 - people.csv: Info about individuals involved
 - vehicles.csv: Info about vehicles involved
- **Output Requirements:**
 - A 20-page PDF report (text + figures)
 - Well-commented Python notebooks (4 tasks in separate folders)

Task 1: Data Understanding & Preparation (30 points)

Task 1.1: Data Understanding

- **Mandatory:** Analyze crashes.csv
- **Recommended:** Use people.csv and/or vehicles.csv
- **Activities:**
 - Data quality assessment
 - Distribution analysis
 - Pairwise correlations
 - Concise data understanding report

Task 1.2: Data Preparation

- **Activities:**
 - Clean and improve data quality
 - Feature extraction (new indicators)
 - Create **monthly profiles for each police department (BEAT_OF_OCCURRENCE)**
- **Suggested Indicators:**
 - Average age of involved people per month/department

- Avg. speed limit per department vs. city average
- Injury level estimates
- **Your Task:** Define **custom indicators** (MANDATORY)
 - Include descriptions and formulas
 - Use in later clustering task
- **Statistical Exploration:**
 - Distributions, outliers
 - Visualizations (Plotly suggested)
 - Correlations
 - Eliminate redundant variables
- **Required Subtasks:**
 - Define semantics for all new/existing features
 - Assess data quality (missing, duplicates, errors)
 - Apply needed transformations



Task 2: Clustering Analysis (30 points + 2 optional)

Objective:

Use extracted features to analyze monthly department profiles with **clustering techniques**.



Subtasks:

1. **K-Means Clustering**
 - Find best k (e.g., elbow method, silhouette score)
 - Describe clusters using centroids and variable distributions
 - Evaluate clustering performance
2. **Density-Based Clustering (e.g., DBSCAN)**
 - Apply to incidents in **one state**
 - Tune parameters
 - Interpret results
3. **Final Comparison:** Evaluate which clustering approach is best
4. **Optional (2 pts):**
 - Try clustering methods from pyclustering library

Task 3: Predictive Analysis (32 points + 2 optional)

Objective:

Predict **DAMAGE** for monthly department profiles

Required Steps:

1. Create target variable:

- Aggregate DAMAGE values per department, year, and month

2. Feature Engineering:

- Review if clustering features are suitable for prediction
- Create new features if needed

3. Model Building:

- Compare models
- Preprocess data if needed
- Split into training and test sets
- Evaluate performance

Mandatory Models:

- Decision Tree
- K-Nearest Neighbors (KNN)
- Ensemble Methods

Optional (2 pts):

- Try other classifiers (e.g., SVM, XGBoost, Logistic Regression)

IMPROVEMENT NEEDS

TASK 1: Data Understanding & Preparation (30 points)

What You Did Well:

- Data overview is informative and comprehensive.
- Variable distributions are analyzed (speed limits, weather, lighting).
- Discussed missing values with appropriate imputation techniques.
- Noted outliers and basic correlation insights.

What You Missed or Lacked:

Requirement from Guidelines	What's Missing / Needs Fixing
Monthly department profiles (<code>BEAT_OF_OCCURRENCE</code> across months/years)	 Not done explicitly; aggregation appears at police beat level but not across time .
Definition of indicators	 No clear mathematical formulation or labeled indicator list. E.g., severity index is used but not formally defined or justified.
Use of at least one other file (<code>people.csv</code> , <code>vehicles.csv</code>)	 Merged files for prediction,  but not for clustering or profiling.
Visualizations (e.g., with Plotly)	 Not included or referenced. Visual aids like maps, heatmaps, or histograms are essential.
Pairwise correlations	 Discussed,  but not supported with a correlation matrix/plot.
Variable transformations and redundancy checks	 Not covered explicitly.
Semantics for each feature	 Not documented. No data dictionary-style section.

Suggestions:

- Create time-series-based profiles per department (`BEAT_OF_OCCURRENCE`) *per month and year*.
- Define 3–5 **key indicators** mathematically.
- Use people and vehicles data earlier (not just for prediction).
- Include **scatter plots**, **heatmaps**, or **geographic visualizations** with Plotly.

TASK 2: Clustering Analysis (30 points + 2 optional)

What You Did Well:

- Applied **K-Means** with normalization and elbow method to find optimal k (4).
- Used **Hierarchical Clustering** and validated 4 clusters using dendrogram.
- **PCA** used for 2D visualization of clusters.

✗ What You Missed or Lacked:

Requirement	Issue
Clustering on monthly profiles	✗ Clustering is done on <i>static beat profiles</i> , not month-by-month department profiles . This breaks a key project requirement.
Cluster characterization (centroids, stats)	✗ No detailed statistical breakdown of each cluster (e.g., avg injuries per cluster).
Comparison of K-means vs Density-based clustering	✗ No DBSCAN or any density-based method was applied.
Optional task: PyClustering library	✗ Not attempted.

🔧 Suggestions:

- Redo clustering on monthly beat profiles (e.g., Jan 2021, Feb 2021, etc.).
- Use DBSCAN or HDBSCAN for density-based clustering on one state subset.
- Include **tables or radar plots** to show how clusters differ.
- Try optional methods from [pyclustering](#) for bonus points.

📌 TASK 3: Predictive Analysis (32 points + 2 optional)

✅ What You Did Well:

- Defined a regression target: average damage per crash.
- Used Decision Tree, KNN, Random Forest with MSE evaluation.
- Feature engineering involved merging datasets.

✗ What You Missed or Lacked:

Requirement	Issue
Prediction at department-month level	✗ Target should be aggregated DAMAGE per department per month/year . You predicted per crash.
Reassessment of features used for clustering	✗ No discussion about whether clustering features were reused or if new ones were created.

Classification vs Regression

✗ Project calls for **classification** (predicting severity bins or damage levels), not just regression.

Alternative models

✗ No exploration of SVM, XGBoost, etc. (Optional 2 pts).

Suggestions:

- Aggregate **DAMAGE** by department–month; define classification bins (e.g., Low, Medium, High).
- Justify which features are reused from clustering and which are new.
- Compare performance across **classification models** (Decision Tree Classifier, KNN Classifier, Random Forest Classifier).
- Discuss confusion matrix, accuracy, F1-score—not just MSE.